# The life-cycle comparison bureau

*Teacher guide and worked answers*

**Compare named animal life cycles using similarities, differences and a valid counterexample.**

40 minutes. Use Lesson.html for interactive teaching or Lesson.pptx for editable slides. Slides.pdf is the static alternative. Select one pupil response route and print the shared evidence it requires. Detailed answers below are for staff.

## Scheme of work

_passsg/inputs/GROW SOW 2026-27.xlsx | GROW Weekly - Spring | C38

Compare the life cycles of different living things.

Intentional deeper comparison after grow_life_cycle_archive; use new evidence and questions. Original secondary-age exemplar at an upper-KS2 conceptual level; no qualification approval or completion claim.

## Prior learning

Describe a named animal life cycle.

Know that adults can reproduce and offspring grow.

## Success evidence

Compare two species using the same feature.

Use a life-cycle stage as evidence for a difference.

Explain why one shared feature does not make two animals the same group.

## Equipment

Shared species cards, selected route, pencils; optional desk sorting labels.

## Preparation

Read the named-species cards; no observations of live animals are required.

Explain that the diagrams omit exact times and are not to scale.

## Practical care

Use the paper archive; do not collect eggs, disturb nests or handle wildlife.

## Ready to teach alternative

All evidence needed is on paper or screen; sort or annotate cards at a desk.

# Teaching sequence

| Slide | Stage | Minutes | Focus |
|---|---|---|---|
| 1 | Opening | 0-2 | Four cases, one comparison challenge |
| 2 | Retrieval | 2-5 | Use the stage words carefully |
| 3 | I do | 5-13 | I choose a feature, then compare |

| Slide | Stage | Minutes | Focus |
|---|---|---|---|
| 4 | I do | Within stage | I use the same feature for the monarch |
| 5 | I do | Within stage | I test an “all” claim |
| 6 | We do | 13-21 | Comparison bureau: shared-feature sort |
| 7 | We do | Within stage | Build a precise comparison |
| 8 | We do | Within stage | The claim challenge |
| 9 | Hinge | 21-24 | Which comparison uses evidence fairly? |
| 10 | You do | 24-34 | Your independent investigation |
| 11 | You do | Within stage | Use the evidence |
| 12 | You do | Within stage | Check before sharing |
| 13 | Review | 34-38 | Compare and improve |
| 14 | Review | Within stage | A question worth investigating |
| 15 | Exit | 38-40 | Show the science you can explain |

Times are a suggested 40-minute route. Slides marked Within stage share the preceding stage allocation. The optional HTML timer starts only when selected.

## 1 Opening Four cases, one comparison challenge

Invite a tentative response. No need to know the platypus in advance: the shared card supplies the needed evidence.

## 2 Retrieval Use the stage words carefully

R1 reproduce; R2 monarch, common frog or another accurate example; R3 no, the next generation starts with new offspring.

## 3 I do I choose a feature, then compare

Think aloud: I compare the presence of a pupa, not whether the animal is attractive or large. The frog route includes a larva and major change, but no pupa. Time spans and detailed anatomy are omitted.

## 4 I do I use the same feature for the monarch

Shared I do time. Think aloud: I use a matching question: Is a pupa present? For the monarch, yes; for common frog, no. This is a justified difference. Do not generalise the monarch route to every insect.

## 5 I do I test an “all” claim

Shared I do time. Explain a counterexample as one real case that breaks the rule. Mammal classification does not depend solely on live birth. Compare named species; do not infer every detail about all mammals from one.

## 6 We do Comparison bureau: shared-feature sort

First group by pupa: M yes; F/P/R no. Then milk feeding: P yes; F/M/R no. All four share egg-laying. Accept desk sorting, pointing or writing.

## 7 We do Build a precise comparison

Both lay eggs and have larvae; monarch has pupa whereas frog does not. Do not accept merely different colours or sizes unsupported by cards.

## 8 We do The claim challenge

Platypus, frog or monarch disproves claim; platypus specifically demonstrates an egg-laying mammal. Four examples do not describe all species. Reveal after pupil reasoning.

Reveal: The platypus lays eggs and is a mammal. That one case breaks the “all birds” claim.

## 9 Hinge Which comparison uses evidence fairly?

A false identical cycles. B correct feature-based comparison. C false all mammals live birth.

A. Frog and monarch have identical life cycles because both lay eggs. - Shared eggs do not erase differences such as the pupa stage.

B. Frog and monarch both have larvae, but only the monarch has a pupa. - The same stages are compared using both cards.

C. A platypus cannot be a mammal because it lays eggs. - The platypus is an egg-laying mammal and its young receive milk.

## 10 You do Your independent investigation

10 minutes across the three You do slides. Supply shared page 1 and one selected route page. All routes target the stated science objective; a labelled drawing or independent dictated explanation is valid.

## 11 You do Use the evidence

Shared You do time. Ask pupils which evidence supports their explanation. Read prompts or scribe without supplying the scientific conclusion.

## 12 You do Check before sharing

Shared You do time. Use the route-specific worked answers to identify one precise next step.

## 13 Review Compare and improve

4 minutes across Review. Offer partner or private teacher feedback. Ask for one specific revision linked to evidence; no public ranking.

## 14 Review A question worth investigating

Lundy cycle: invite written, spoken or pointed suggestions. Select one feasible question and explain how the pupil contribution changes the next example or investigation.

## 15 Exit Show the science you can explain

Eggs/larvae shared; pupa only monarch; any named non-bird egg-layer with accurate reason. E4 on page asks about the next generation.

# Answers and assessment

## R1-R3

Reproduce; a named animal with a larva such as frog or monarch; no, the next generation contains new individuals.

### W1-W3

W1 pupa: monarch yes, other three no; milk: platypus yes, other three no. Groups change because the feature changes. W2 both eggs/larvae; monarch has pupa and frog does not. W3 frog, monarch or platypus is a non-bird egg-layer; platypus is a mammal.

### Hinge A-C

A false: same feature is not identical cycle. B correct. C false: platypus is an egg-laying mammal.

### S1-S5

S1 egg. S2 larva; pupa. S3 egg → caterpillar → pupa → adult. S4 no, mammal. S5 name a non-bird egg-layer and state that this contradicts all egg-layers being birds.

### N1-N4

N1 frog no/no; monarch yes/no; platypus no/yes; robin no/no. N2 both eggs or larvae, but pupa only monarch. N3 both lay eggs; platypus young receive milk whereas robin chicks receive other food from parents. N4 valid counterexample; a sample of four cannot establish every possible animal life cycle.

### T1-T4

T1 three accurately matched features, for example frog and monarch: egg location pond versus milkweed, larvae both, pupa absent versus present. T2 false: tadpoles are larvae but no pupa follows. T3 false: platypus lays eggs and feeds young milk. T4 missing information is not evidence of absence; record not supplied/unknown and seek a reliable source.

### E1-E4 / assessment

E1 eggs or larvae. E2 pupa only monarch. E3 named correct non-bird case and contradiction. E4 adults produce new offspring, not themselves becoming eggs. Secure: matched comparison and valid counterexample; re-teach vague differences by selecting one common feature question. Reflection accepts a relevant evidence question.

### Teaching and assessment notes

40 minutes: opening 2; retrieval 3; I do 8; We do 8; hinge 3; You do 10; review 4; exit 2. Zero-minute slides share their stage time.

Supply shared page 1, one selected route page 2, 3 or 4, and page 5. Do not require all three routes. An independent spoken explanation can be recorded verbatim by an adult.

The comparison extends the earlier life-cycle archive through new comparison questions and a matched-feature matrix. It intentionally revisits species knowledge rather than claiming untouched content.

Interactive sorting decisions map to W1-W3; paper cards support the same predict, check and explain cycle.

## Access and responsive teaching

### Supported

Compare two named species with a feature word bank.

### Standard

Complete a feature table and justify one similarity and two differences.

### Stretch

Test broad classification claims and state the limits of the sample.

### Regulation

Offer private writing, pointing, drawing or adult scribing. Allow pass-and-return for public questions and desk-based participation. Choose support from current learning evidence, not fixed ability labels.

## Misconceptions to check

**All animals that lay eggs are birds.**

Many other animals lay eggs; a platypus is an egg-laying mammal.

Check: Ask which card provides a counterexample.

**A pupa is just another word for any baby animal.**

A pupa is a particular stage in insects such as monarch butterflies.

Check: Ask whether a frog has a pupa.

**An adult turns back into its own egg.**

Adults produce a new generation; an individual does not become an egg again.

Check: Ask whose life starts at the next egg.

| Word | Meaning |
| --- | --- |
| compare | Describe similarities and differences using the same feature. |
| larva | An immature stage with a different body form from the adult. |
| pupa | A developmental stage between larva and adult in insects with complete metamorphosis. |
| mammal | An animal in the group whose females produce milk for young. |
| counterexample | A case that shows an all-or-never claim is false. |

## Scientific references

Froglife: Common frog

Checked September 2026. Common-frog eggs, tadpoles and adult life history. Activities and diagrams are original.

Monarch Joint Venture: Life cycle

Checked September 2026. Named monarch sequence: egg, larva, pupa, adult; not a rule for all insects.

Australian Museum: Platypus

Checked September 2026. Egg-laying mammal; young receive milk. This is a counterexample to all mammals giving live birth.

RSPB: Robin

Checked September 2026. Eggs, nest and young robin development; no nest disturbance is needed.